1 Periodic Functions and Their Graphs

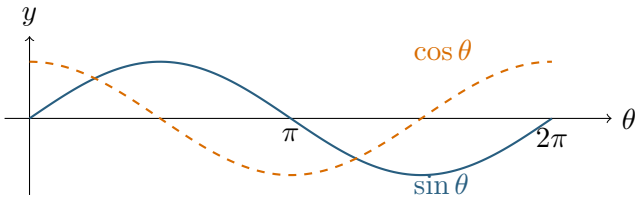
Central question. How does rotation become a repeating signal, and how can we recover the motion from a graph?

1.1 The circle makes a wave

For a point traveling around the unit circle with angle θ, its vertical coordinate is $y = \sin\theta$ and its horizontal coordinate is $x = \cos\theta$. Each revolution restores the same coordinates. Hence

$$\sin(\theta + 2\pi) = \sin\theta, \qquad \cos(\theta + 2\pi) = \cos\theta.$$

The *period* is the smallest positive input change that repeats the full function. Both sine and cosine have period 2π. Sine starts at 0 and initially rises; cosine starts at 1.



1.2 Amplitude and vertical translation

Consider $y = A\sin x + D$. Its midline is $y = D$, its amplitude is $|A|$, and its range is $[D - |A|, D + |A|]$. A negative A reflects the wave across the midline.

Example 1.1. The function $y = 3\sin x - 2$ has amplitude 3, midline $y = -2$, maximum 1, and minimum -5.

1.3 Period and horizontal scaling

For $y = \sin(Bx)$ with $B \neq 0$, one full cycle occurs when Bx changes by 2π; the period is $T = 2\pi/|B|$.

Example 1.2. The graph of $y = \cos(4x)$ repeats every $\pi/2$ units of input.

When x denotes time measured in seconds, the quantity $|B|$ is angular frequency in radians per second, and ordinary frequency is $f = 1/T = |B|/(2\pi)$ cycles per second.

1.4 Phase shift

A useful general form is

$$\boxed{y = A\sin(B(x - C)) + D}.$$

Here C is the horizontal shift, D the midline, $|A|$ the amplitude, and $2\pi/|B|$ the period. We should distinguish $B(x - C)$ from $Bx - C$: factoring first prevents an incorrect phase shift.

Example 1.3. For $y = 2\sin(3x - \pi) + 1 = 2\sin(3(x - \pi/3)) + 1$, the period is $2\pi/3$, the shift is $\pi/3$ to the right, the amplitude is 2, and the midline is 1.

1.5 Five landmark points

Rather than sketching a sinusoid by guesswork, divide one period into four equal intervals. For $A > 0$, sine begins on the midline, rises to a maximum after one quarter-period, returns to the midline, reaches a minimum, then returns to the midline. A negative amplitude reverses the rises and falls.

Try it. Find five landmark points of $y = -2\cos(\pi(x-1)) + 3$, showing one entire period.

1.6 Models of repeating motion

A Ferris wheel has radius 9 m, its center is 11 m above ground, and it makes one revolution every 30 s. If a passenger begins at the rightmost position and travels counterclockwise, their height is

$$h(t) = 11 + 9\sin\left(\frac{2\pi}{30}t\right).$$

Its amplitude is 9 m, midline is 11 m, and period is 30 s. A sinusoidal model can describe a wheel, a spring, a tide, or a musical tone, provided its parameters actually correspond to the chosen physical quantities.

1.7 Tangent and reciprocal graphs

The tangent function repeats after π, not 2π, because both sine and cosine change sign after a half turn and their ratio remains the same:

$$\tan(x+\pi) = \tan x.$$

Tangent has vertical asymptotes at $x = \pi/2 + k\pi$. The secant and cosecant graphs arise as reciprocals of cosine and sine; their breaks occur wherever the respective denominator is zero.

1.8 Exercises

1. Find the amplitude, midline, period, and range of $y = -4\sin(2x) + 1$.

2. Rewrite $y = 3\cos(5x - \pi) + 2$ in shifted form and identify the phase shift.

3. Give five landmark points for $y = 2\sin(x - \pi/2)$.

4. Explain why $y = \sin(2x)$ completes two periods on $[0, 2\pi]$.

5. A wheel of radius 6 m rotates once every 24 seconds. Its center is 8 m above the ground. Model the rider's height when starting at the rightmost point.

6. Identify the vertical asymptotes of $y = \tan(2x)$ on $[0, 2\pi)$.

7. Sketch one period of $y = 2\sec x$ and indicate all discontinuities.

8. Explain how you can determine amplitude and midline directly from observed maximum and minimum values.